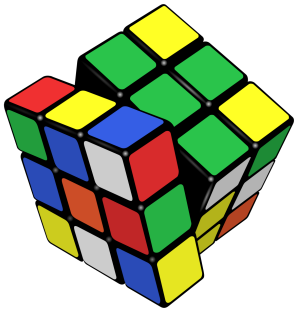
Student Name:

Parabola Transformations and Group Theory Extra Credit



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1 Introduction and Background

When mathematicians use the word "Algebra", they usually aren't talking about the types of things you'd learn in Algebra I, Algebra II, or Precalculus like solving and graphing equations, evaluating functions, manipulating equations, parabolas, trigonometry, etc. Instead they are referencing the field of Abstract Algebra. Abstract algebra is the study of mathematical objects. The main idea is that if you can define rules for how an object behaves, anything that follows those rules *is* that object! And whatever you prove about the general object is also true in your specific case. A very basic example of a mathematical object is a shape. If we wanted to calculate the volume of a golf ball, a baseball, and a basketball, you wouldn't need separate formulas for each one. You'd just say "All of these are spheres, so I can use the formula for the volume of a sphere". That is the crucial step of abstraction, and it is what gives math its power. If we can show that something behaves like a thing we already know a lot about, we can just apply that knowledge instead of coming up something new.

One of the most beautiful (and useful) mathematical objects is the group. Groups have to do with symmetry and transformations. They show up all over the place in science, including chemistry, physics, and biology. The main idea with groups is this: If you can represent a certain symmetry with a certain group, then you can just work with the (usually simpler) abstract group and get the exact same result. Or, if you show that two separate objects represent the same group, you can switch between the two objects and get the same result.

A group has two parts: a set of elements, and an operation that combines those elements. They follow 4 rules called "Axioms" that we will discuss later.
Some classic examples of groups are:

- $\mathbf{SO}(3)$
  - represents rotations in 3-D space. Elements are the rotations, operation is doing one rotation and then another. This is extremely useful in physics. Allowing for "rotations" in time gives you $\mathbf{SO}(1,3)$, the lorentz group, which is the fundamental group of special relativity, the theory that tells us there is a finite speed of light, and $E = mc^2$
- $\mathbb{Z}_n$
  - cyclic group of order n, can represent rotational symmetry of a regular polygon with n sides. Elements are whole numbers 0 to $n-1$, operation is addition $\mod n$ (normal addition, but at the end you divide by n, and your answer is the remainder)
- $\mathbf{T}(2)$
  - represents translations in 2-D space. Similar to $\mathbf{SO}(3)$, elements are the translations, and the operation is doing one and then the other. This will be especially useful for us, since h and k translate the parabola in 2 dimensions.

One of the main ideas we have learned about quadratics and parabolas is the transformations of a parabola when it is in vertex form. As a reminder, vertex form looks like this:

$$y = a(x-h)^2 + k \tag{1}$$

Each parameter (a, h, and k) controls a different transformation of the parent function $y = x^2$. a will vertically stretch, compress, and flip the parabola. h will translate the parabola horizontally, and k will translate the parabola vertically. The goal of this assignment is to create a group that represents these transformations. Our plan of action will be this: show that the transformations corresponding to a, h, and k are groups by themselves, then show that when put together we still have a group. We will also try to find a new representation of this group that is easier to work with than vertex form quadratics.

Problem 1

In order to come up with a group for transformations of a parabola, we need to decide what our elements and operation are. The way to do this that makes the most sense to me is defining our elements as the transformations to get from the parent function $y = x^2$ to each parabola $y = a(x-h)^2 + k$, and defining our operation as performing these transformations one after another (composition)

(a) write the equation of a parabola in vertex form corresponding to the transformation "translate to the right by 2". From now on we will call this transformation $H(2)$. What does $H(n)$ do to h?

(b) write the equation of a parabola in vertex form corresponding to the transformation "translate up by 3". From now on we will call this transformation $K(3)$. What does $K(n)$ do to k?

(c) write the equation of a parabola in vertex form corresponding to the transformation "vertical stretch of 5". From now on we will call this transformation $A(5)$. What does $A(n)$ do to a?

(d) write the equation of a parabola in vertex form with all three of these transformations. This would correspond to the transformation
$A(5) \cdot K(3) \cdot H(2)$

(e) does the order you perform the transformations matter? why?

When the order of operations for group elements doesn't matter, that is $A \cdot B = B \cdot A$ for all A, B in the group. We call that group **Abelian**, and say the elements **commute**. What you have just done in problem 1, is create the beginnings of our parabola transformation group, and show that it is an Abelian group. It also hinted that we may be able to shed the baggage of working with vertex form, and just look at what these transformations do to a, h, and k

2 Definition of a Group

A group $\mathbf{G} = (S, \cdot)$ is a set of elements S and an operation $\cdot$ that satisfy the following rules:

1. Closure: For all A, B in the group, $A \cdot B$ is also in the group.
2. Associativity: For all A, B, C in the group, $(A \cdot B) \cdot C = A \cdot (B \cdot C)$.
3. Identity element: There exists an element I in the group such that for all A in the group, $I \cdot A = A \cdot I = A$.
4. Inverse element: For every A in the group, there exists an element A^{-1} in the group such that $A \cdot A^{-1} = A^{-1} \cdot A = I$.

Let's think about this conceptually. As long as your elements and operation follow these rules, they are a group! Your elements could be just about anything: numbers, functions, shapes, graphs, vectors, transformations, spacetime itself etc. Similarly, there is no restriction on what your operation is. It could be addition, multiplication, exponentiation, performing one action and then another, basically any way to combine two elements of your group. In the next 4 problems, you will prove that our transformation group for parabolas follows each of the group axioms, and is therefore a group!

Problem 2
What is closure? Well, closure just says that when we apply our operation to combine two elements of the group, the result must also be in the group. For Example: if our elements are the positive integers $\mathbb{Z}^+$ (1,2,3,4,5...), then they would be closed under addition, since any positive integer plus another positive integer is also a positive integer. However, they are NOT closed under subtraction, because a positive integer minus a positive integer could be zero or negative (for example $5 - 7 = -2$), which are not positive integers.

(a) show that the positive integers are closed under multiplication

(b) show that the positive integers are NOT closed under division. (Find a counterexample)

(c) show that horizontal translations of a parabola $H(n)$ are closed for all real numbers n under composition. That is, show that no matter which two translations we combine (like this: $H(a) \cdot H(b) = H(a+b)$, we will always get another valid translation of a parabola. (Hint: use vertex form)

(d) does this result also apply to vertical translations K(n)? Why?

(e) in problem 1c, you saw that vertical stretching could be represented by the transformation $A(n)$, where n represented the amount of stretch. If we use numbers less than 1, we can do a compression, for example: $A(0.25)$. If we use negative numbers, we can flip the parabola, for example $A(-1)$. If we try $A(0)$ however, we have an issue. We saw that $A(n)$ multiplies the a of a parabola by n. If we try that with $A(0)$, we would get $y = 0(x-h)^2 + k = k$, which is a horizontal line, not a parabola. This would violate closure. We can deal with this by excluding $A(0)$ from our group. Show that the stretching, compressing, and flipping transformations $A(n)$, where n is a nonzero real number, are closed under composition. Hint: $A(a) \cdot A(b) = A(ab)$

Problem 3

The next group axiom is associativity. You've probably heard of associativity before via the associative property. Associativity is very common. Almost every interesting algebraic structure has associativity (the ones without are called magmas if you're curious). To prove associativity for our group, we just have to show that its elements and operation follow the associative property $(A \cdot B) \cdot C = A \cdot (B \cdot C)$. To prove this, you can literally just evaluate the left and right sides of that equation. I'll show how to do this for $K(n)$, and you will do it for $H(n)$ and $A(n)$.

We want to show, with real numbers l, m, and n:

$$(K(l) \cdot K(m)) \cdot K(n) = K(l) \cdot (K(m) \cdot K(n)) \tag{2}$$

In words, the left hand side would be equivalent to "translate vertically by n, then translate vertically by $l + m$". You can see that this is equivalent to translating by $m + n$ then l, which corresponds to the right hand side. We can also show this symbolically. We saw earlier that our composition operation acts on $K(n)$ like this: $K(a) \cdot K(b) = K(a + b)$. Using this we have:

$$(K(l) \cdot K(m)) \cdot K(n) = K(l + m) \cdot K(n) \tag{3}$$

$$K(l + m) \cdot K(n) = K((l + m) + n) \tag{4}$$

Then, by the associative property of addition:

$$K((l + m) + n) = K(l + (m + n)) \tag{5}$$

Expanding the right side back into 3 transformations we get:

$$K(l + (m + n)) = K(l) \cdot K(m + n) = K(l) \cdot (K(m) \cdot K(n)) \tag{6}$$

which proves that the vertical translations of a parabola $K(n)$ are associative. Another way to look at this is by applying each side of equation 2 to actual parabolas. If we apply them to an arbitrary parabola $y = a(x - h)^2 + k$ The left hand side would give:

$$y = a(x - h)^2 + k + (l + m) + n \tag{7}$$

and the right would give:

$$y = a(x - h)^2 + k + l + (m + n) \tag{8}$$

These are equal by the associative property of addition. Ok, your turn:

(a) show symbolically that the horizontal translations of a parabola $H(n)$ are associative. Also show what it would look like to apply each side of the associativity equation to an arbitrary parabola

(b) show symbolically that the stretching/compression/flipping transformations of a parabola $A(n)$ are associative. Also show what it would look like to apply each side of the associativity equation to an arbitrary parabola. Remember that multiplication is also associative over the real numbers!

Problem 4

The identity is often the most boring, and simultaneously the most important element in any group. Without it, you wouldn't have a group! Conceptually, the identity is a transformation that does nothing. When composed with another element of the group, you just get that element back. Symbolically that looks like this for our parabola groups:

$$I \cdot K(n) = K(n) \tag{9}$$

$$I \cdot H(n) = H(n) \tag{10}$$

$$I \cdot A(n) = A(n) \tag{11}$$

Let's find the identity for each of these groups of transformations.

(a) Find the identity element for $H(n)$. That is, what value of n would create a transformation that does nothing to a parabola?

(b) Find the identity element for $K(n)$

(c) Find the identity element for $A(n)$ (This one will be different!)

Problem 5

Inverse elements "undo" whatever transformation a certain element does. For example, the inverse of a translation to the right by 17 would be a translation to the left by 17. Symbolically we write the inverse of an element A like this: A^{-1}, referencing good old multiplicative inverses. When multiplying numbers by their inverse we get: $x \cdot x^{-1} = 1$. With elements of a group, we want the composition of inverses to be a transformation that does nothing, i.e. the identity: $A \cdot A^{-1} = I$. For the inverse axiom to be satisfied, **every** element must have an inverse. Since we now know the identity for each of our parabola transformation groups, we can find a general form of an inverse for each one.

(a) what is the inverse of the transformation $H(2)$?

(b) what is the inverse $H(n)^{-1}$ of an arbitrary horizontal translation $H(n)$? That is, find $H(?)$ such that $H(n) \cdot H(?) = I$

(c) does this work for any real n?

(d) what is the inverse $K(n)^{-1}$ of an arbitrary vertical translation $K(n)$? That is, find $K(?)$ such that $K(n) \cdot K(?) = I$

(e) what is the inverse $A(n)^{-1}$ of an arbitrary stretch/compression/flip $A(n)$? That is, find $A(?)$ such that $A(n) \cdot A(?) = I$. remember, $n \neq 0$

(f) for each of these groups, what is the inverse of the identity?

Ok, so now we have shown that the horizontal translations, vertical translations, and stretching/compressing/flipping transformations form groups under composition. We would write the definitions of these groups symbolically like this (the $\in$ symbol just means "is a member of" and $\mathbb{R}$ just means the real numbers):
Group **H** of Horizontal Translations:

$$\mathbf{H} = (H(n), \cdot), n \in \mathbb{R} \tag{12}$$

$$H(a) \cdot H(b) = H(a + b) \tag{13}$$

Group **K** of Vertical Translations:

$$\mathbf{K} = (K(n), \cdot), n \in \mathbb{R} \tag{14}$$

$$K(a) \cdot K(b) = K(a + b) \tag{15}$$

Group **A** of Stretches, Compression, and Flips:

$$\mathbf{A} = (A(n), \cdot), n \in \mathbb{R} \neq 0 \tag{16}$$

$$A(a) \cdot A(b) = A(ab) \tag{17}$$

3 Combining Groups

using our fancy new groups we can do some cool tricks. Say we start with the parent function parabola $y = x^2$, or equivalently, $y = 1(x - 0)^2 + 0$. We could represent this with the identity for each group, since it has undergone no transformations relative to itself: $A(1)$, $H(0)$, $K(0)$. If we wanted to get to some arbitrary parabola $y = a(x - h)^2 + k$, we could apply the transformations $A(a)$, $H(h)$, and $K(k)$. Lets test it out:

Problem 6

Remembering our rules for composing our transformations (Equations 13,15, and 17) find the resulting transformation, and the corresponding parabola in vertex form for the following:

(a) $A(1) \cdot A(a)$

(b) $H(0) \cdot H(h)$

(c) $K(0) \cdot K(k)$

(d) $K(-2) \cdot K(4)$

But what if we didn't want to work 3 groups at once? is there a way to combine them? Indeed there is! It's called the direct product. It basically puts each group in a list and deals with them separately, but that list becomes a group itself! The direct product $\mathbf{A} \times \mathbf{B}$ of two groups $\mathbf{A} = (A, \cdot)$ and $\mathbf{B} = (B, +)$ is a

group, and is defined as follows:

$$\mathbf{A} \times \mathbf{B} = ((A, B), *) \tag{18}$$

$$(A_1, B_1) * (A_2, B_2) = (A_1 \cdot A_2, B_1 + B_2) \tag{19}$$

Note that the $\cdot$ and $+$ don't necessarily mean multiplication and addition, I'm just showing that $\mathbf{A}$ and $\mathbf{B}$ may have different operations. If $\mathbf{A}$ has identity I_A, and $\mathbf{B}$ has identity I_B, the the identity for $\mathbf{A} \times \mathbf{B}$ is:

$$I_{A \times B} = (I_A, I_B) \tag{20}$$

Similarly, if A has inverse A^{-1} and B has inverse B^{-1}, then

$$(A, B)^{-1} = (A^{-1}, B^{-1}) \tag{21}$$

Note that this only works if the groups have no overlapping elements. Let's do this for our parabola groups.

Problem 7

We have 3 groups, $\mathbf{A}$, $\mathbf{H}$, and $\mathbf{K}$ that we want to combine. I'm going to call the combined group $\mathbf{P}$ (for parabola). We have:

$$\mathbf{P} = \mathbf{A} \times \mathbf{H} \times \mathbf{K} = ((A(a), H(h), K(k)), *) \tag{22}$$

with $a \in \mathbb{R} \neq 0$ and $h, k \in \mathbb{R}$

(a) find the definition of $*$. That is, create something like Equation 19 for our group $\mathbf{P}$. Remember, the elements of $\mathbf{P}$ are lists of elements of $\mathbf{A}$, $\mathbf{H}$, and $\mathbf{K}$

$$(A(a_1), H(h_1), K(k_1)) * (A(a_2), H(h_2), K(k_2)) = ? \tag{23}$$

(b) find the identity for $\mathbf{P}$

(c) find the general form of an inverse for $\mathbf{P}$ (Something like equation 21)

Problem 8

Once we've done the legwork of defining $\mathbf{P}$, we can use our old transformations on it. for example, if we wanted to translate to the right by 3, we could define our old $H(3)$ as $(A(1), H(3), K(0))$, when acting on an element of $\mathbf{P}$. This transformation would do nothing to the a and k of our parabola, but it would add 3 to h, translating it to the right by 3. I'll call writing down the only non identity transformation as "abbreviated form" and writing out the whole list as "complete form". We can compose transformations from $\mathbf{A}$, $\mathbf{H}$, and $\mathbf{K}$ with each other, and write the result in complete form.

(a) write $A(2) \cdot H(-1)$ in complete form. Write the vertex form equation that corresponds to these transformations, and a description of the transformations in words.

(b) write $A(2) \cdot A(-1) \cdot H(3) \cdot K(-2)$ in complete form. Write the vertex form equation that corresponds to these transformations, and a description of the transformations in words.

(c) show that $A(2) \cdot A(-1)$ is equivalent to $(A(2), H(0), K(0)) * (A(-1), H(0), K(0))$

4 Shedding Loose Weight: a New Representation

You may have noticed that we are carrying a lot of letters around. This is not strictly necessary. The capital letters, A, H, and K remind us that we are working with transformations. They can be applied to one another. The lowercase letters tell us how much to translate, stretch, compress, or flip by. If we remember what order the transformations go, we can just write the lowercase letters in complete form. I'm going to call this "vector form"

Problem 9: Proving Vector Form Works

show that the set of objects of the form (a, h, k) with $a \in \mathbb{R} \neq 0$ and $h, k \in \mathbb{R}$, along with the operation * defined as:

$$(a_1, h_1, k_1) * (a_2, h_2, k_2) = (a_1 a_2, h_1 + h_2, k_1 + k_2) \tag{24}$$

form a group by proving the 4 group axioms. Reference problem 2 if you get stuck. Note that you can do this without ever thinking about parabolas, and only taking into account the information in this problem and the group axioms.

(a) Closure

(b) Associativity

(c) Identity

(d) Inverse (Make sure to find the inverse of the identity ;))

With vector form, we can represent any transformation of a parabola with 3 numbers, and combine those transformations with a simple rule! Lets try it out.

Problem 10: Using Vector Form

(a) find $(2, 3, 4) * (0.5, -3, -4)$. Write an equivalent expression using $A(2)$, $H(3)$, etc. Describe what happened using the language of transformations.

(b) find $(1, 3, 5) * (-1, 2, 3) * (-6, 10, 19) * (-0.5, -4, -5)$. Write a vertex form equation corresponding to the resulting transformation.

(c) write a vector form representation of the transformation required to reach the parabola $y = a(x-h)^2 + k$

5 Final Thoughts

You might be wondering, whats the point of all this? Why even create these representations when we could just think about parabolas or equations? Well, by showing that all these objects share the same group structure, we have shown that they are in some sense, all the same. Let's list all the representations we used! Graphs of parabolas, parabolas in vertex form, transformations in words, transformations $A(n)$, $H(n)$, and $K(n)$, complete form, and vector form. That's a lot! Each of them have their own purpose. We can use them and think about them interchangeably because we proved they were all representations of the same underlying group. If I had shown you vector form at the beginning of this assignment, would you have thought it was connected to parabolas? Abstract math can create connections between topics that at first glance seem totally unrelated, and I hope I have given you a taste of that. Another nice thing about the group structure is finding connections with other well known groups. Remember the group of translations in 2D $\mathbf{T}(2)$? The group of 2D translations is just 2 copies of the group of 1D translations in a trenchcoat:

$$\mathbf{T}(2) = \mathbf{T}(1) \times \mathbf{T}(1) \tag{25}$$

And our groups $\mathbf{H}$ and $\mathbf{K}$ are both equal to $\mathbf{T}(1)$ (since they are just 1D translations) so,

$$\mathbf{H} \times \mathbf{K} = \mathbf{T}(2) \tag{26}$$

We can go even further: our group $\mathbf{A}$ (remember this represents stretches, compressions, and flips) can be broken up in to two groups, one for compressions and stretches, and one for flips. The stretch/compression group is called the multiplicative group of the positive real numbers, and it can be written like this: $\mathbf{M} = (\mathbb{R}_{>0}, \times)$. The group for flips only has two elements, 1 and -1 under multiplication. It's called the cyclic group of order 2 or $\mathbb{Z}_2$. I'll let you think about why this is equivalent to the other definition of cyclic groups I gave in the beginning. Putting this all together, we can show the so-called subgroups of our parabola group $\mathbf{P}$:

$$\mathbf{P} = \mathbb{Z}_2 \times \mathbf{M} \times \mathbf{T}(2) \tag{27}$$

Well, if you've made it this far, I'm thoroughly impressed. Please write a short reflection about this assignment in the remaining space. Please include your interpretation of what Equation 27 means. I'm happy to answer any questions you have about group theory, and I'd love any feedback you have about this assignment.